

BIT 4107 – Mobile Application Development

ENHANCED PROFESSIONAL STUDENT LOGBOOK

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Admission Number: BIT/2024/73474

School/Faculty: Computing& Informatics

Programme: Information Technology

Academic Year: 2026

Semester: Two

Lecturer: Michael Nyoro

Phone Number: 0758345174

Instructions

This logbook is used to document weekly learning activities, practical exercises, skills acquired, challenges encountered, solutions applied, and reflections. Students should complete all sections every week.

Each week to follow remarks in week 1 section.

Autogenerated Table of content

Week 1: Introduction to Mobile Application Development

Practical Activities Undertaken

1. Installed Android Studio.
2. Installed Flutter SDK and Dart SDK.
3. Configured Android Studio with Flutter and Dart plugins.
4. Set up Android Emulator.
5. Created the StarShare project.
6. Created a GitHub repository for project version control.
7. Verified Flutter installation and environment configuration.
8. figure1: Installation and setting up of android studio.



Figure2:Configured Android SDK and Emulator and Verified development environment.

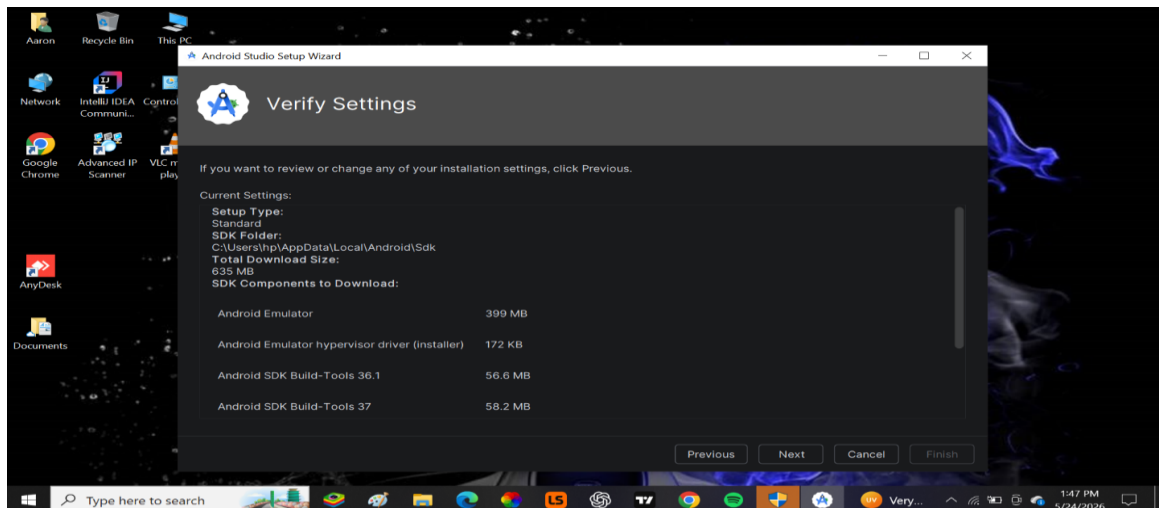
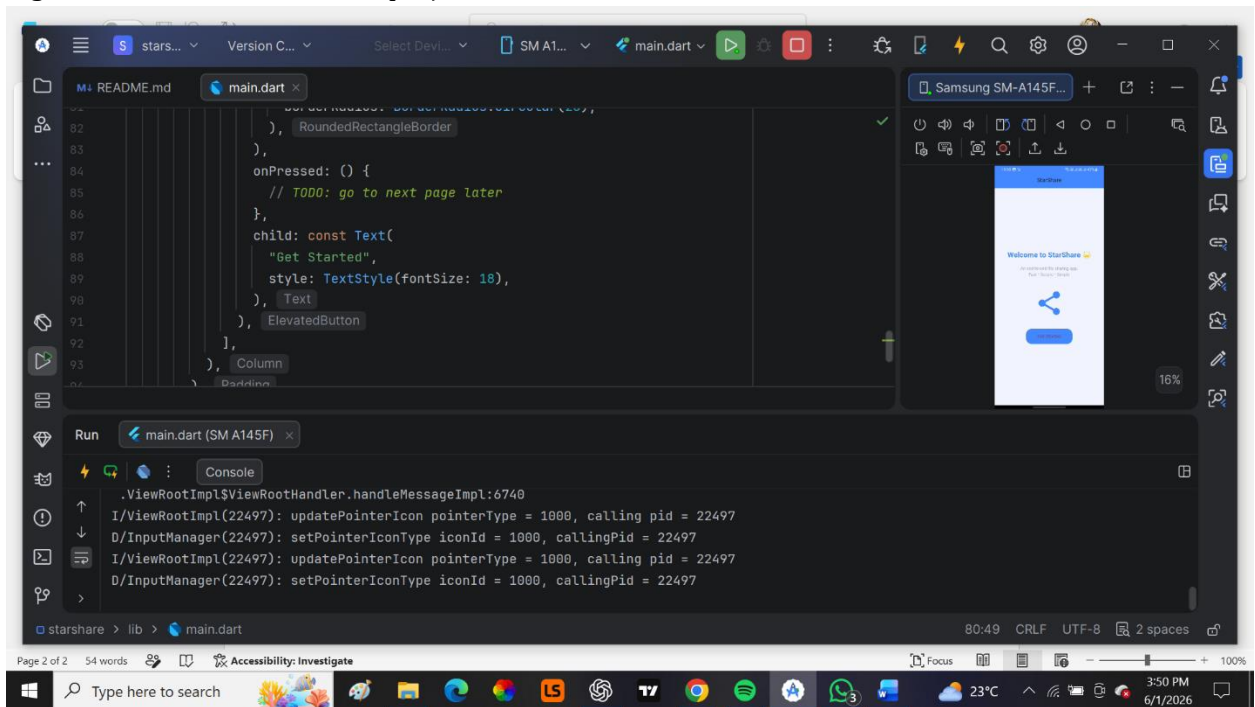
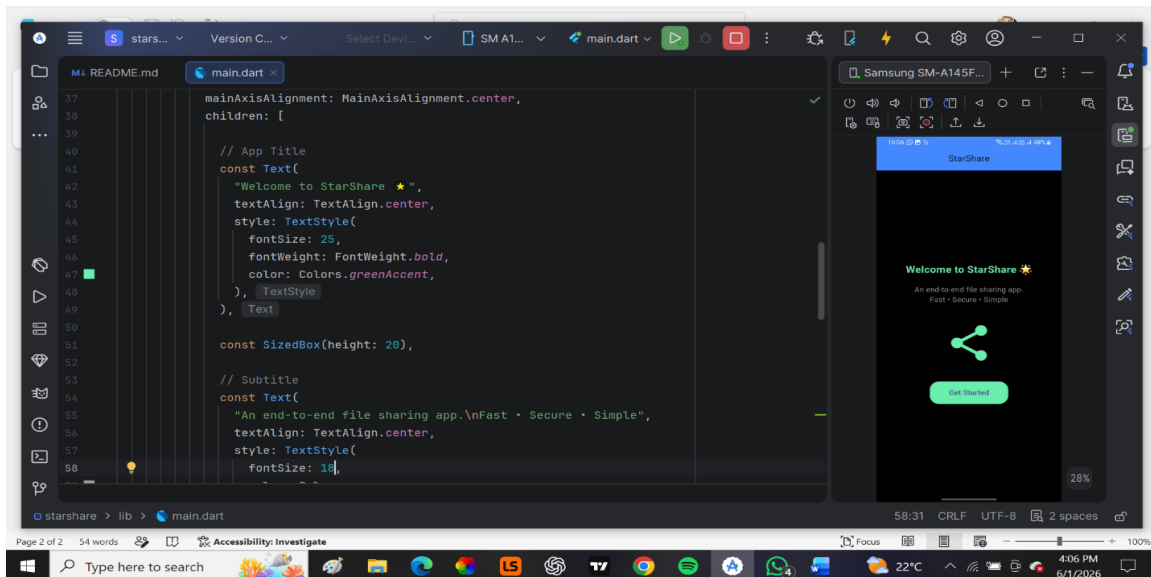


Figure3: Created a new flutter project.



- Figure4: Modified the application title, changed the background colour and adjusted the font size of the application text.



Tools/Software Used

1. Android Studio
2. Flutter SDK
3. Android Emulator
4. Phone

5. Dart SDK
6. Figma

Personal Reflection

I Successfully installed and configured the development environment for the StarShare project. I gained practical experience setting up Flutter, Android Studio, and GitHub, which established a solid foundation for the development process.

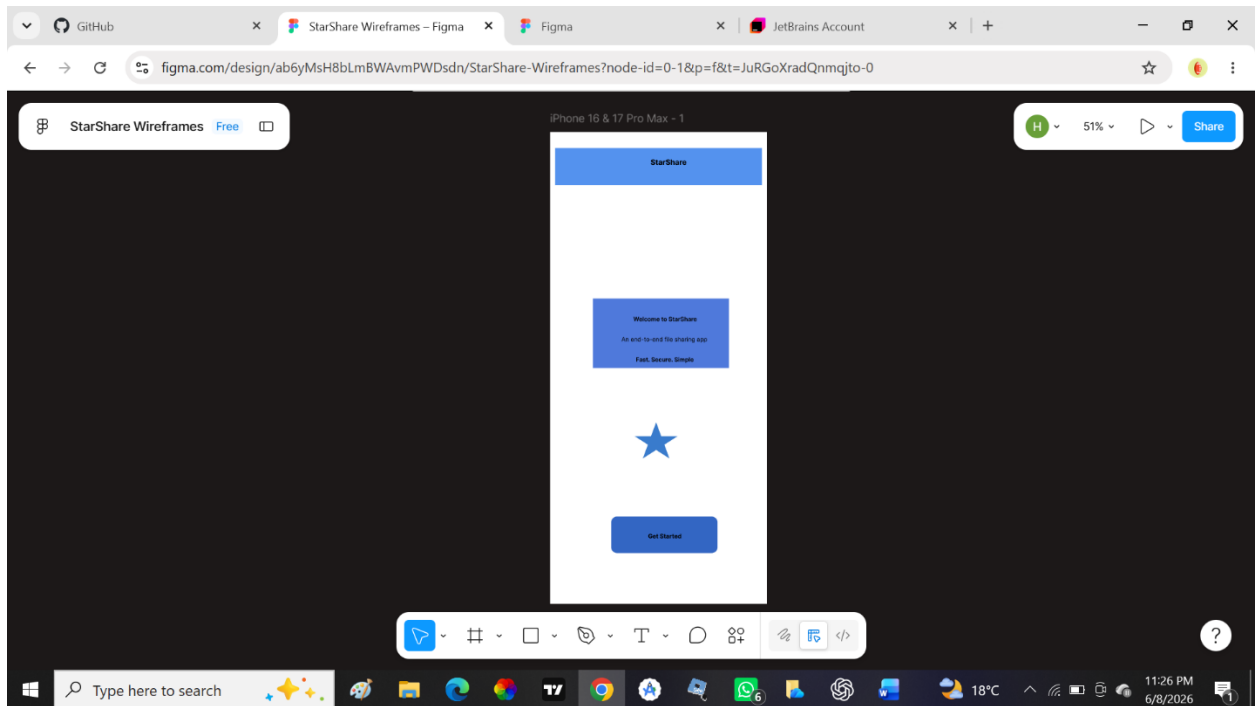
Week 2: Mobile Development Environment

Practical Activities Undertaken

1. Installed and configured Figma.
2. Explored mobile application UI/UX design principles.
3. Created project wireframe layouts in Figma.
4. Designed Splash Screen wireframe.
5. Designed Welcome/Onboarding Screen wireframe.
6. Designed Login Screen wireframe.
7. Designed Registration Screen wireframe.

8. Reviewed mobile application navigation structures.

9. **Figure 1:** Onboarding Screen Wireframe.



10. **Figure 2:** Splash Screen Wireframe.

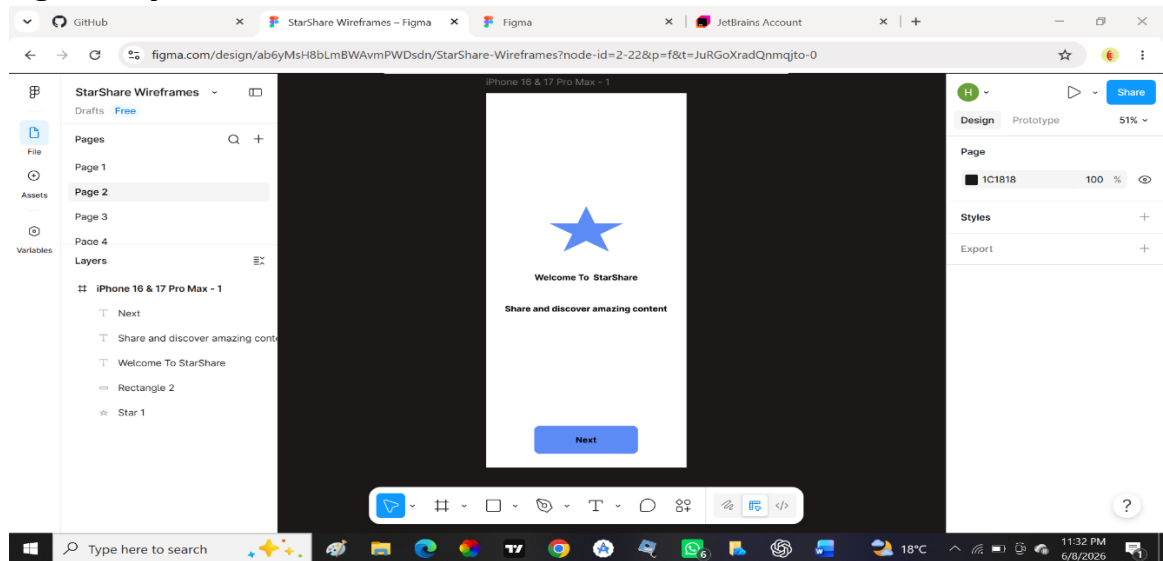


Figure 3: Login screen Wireframe

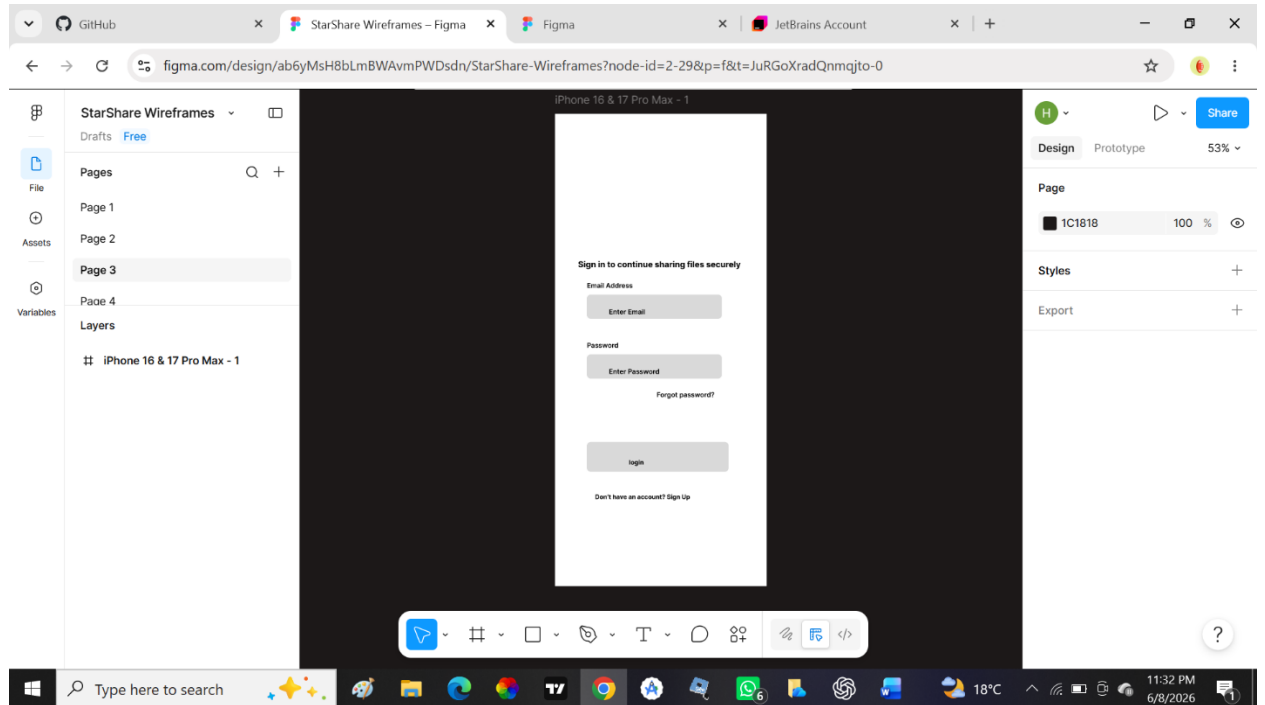
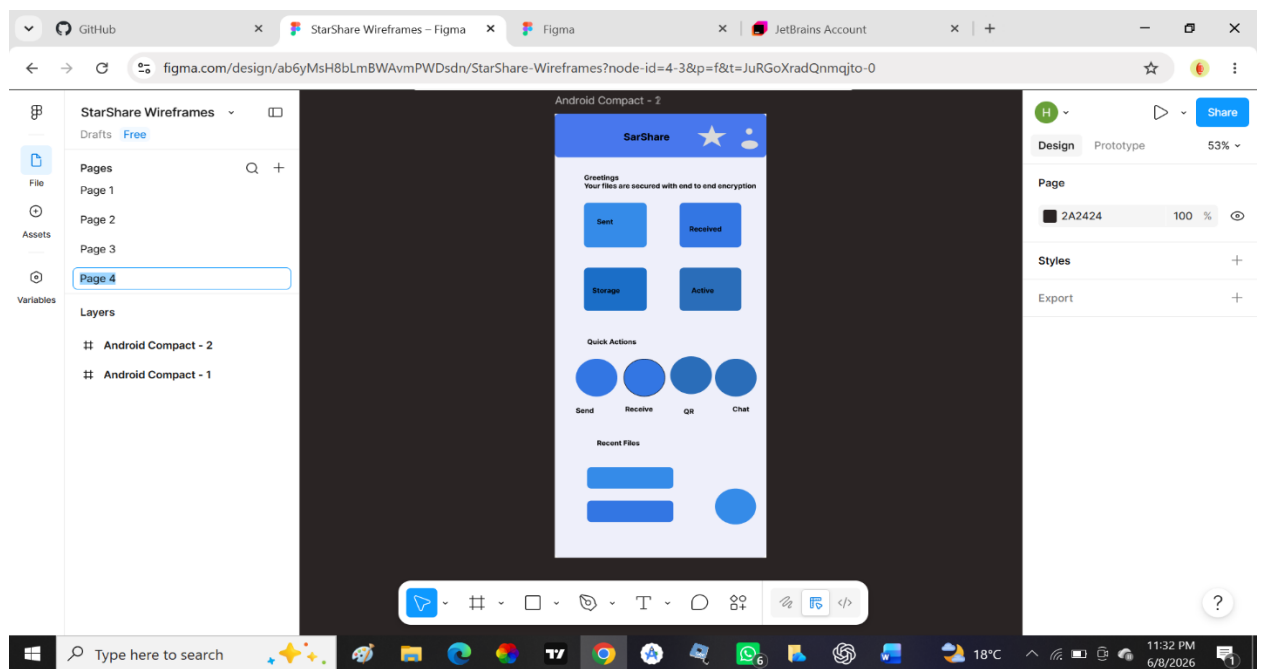


Figure 4: Home Screen Wireframe



Tools used.

- 1.Figma
2. Android studio (for references)

Personal Reflection

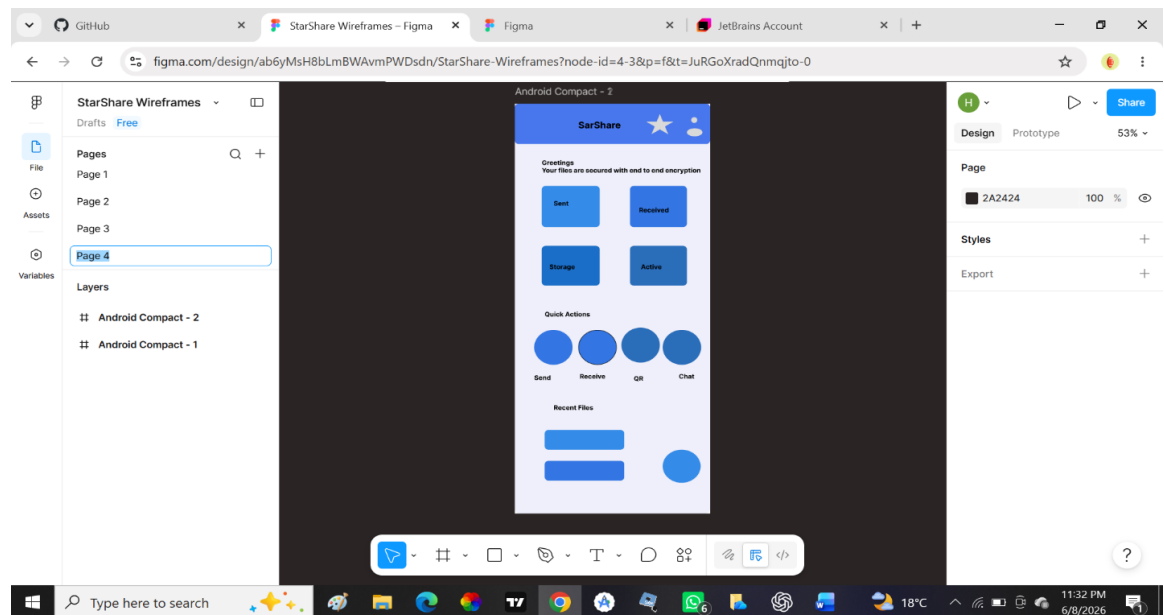
I developed wireframes using Figma and applied UI/UX design principles to improve usability. This activity provided a visual representation of the application structure and guided future development decisions.

Week 3: User Interface Design

Practical Activities Undertaken

1. Refined application wireframes.
2. Designed Home Dashboard interface.
3. Designed File Upload section
4. Designed Storage section
5. Improved navigation flow between screens.
6. Standardized buttons, icons, and layout structures.
7. Reviewed interface consistency and usability.

Figure 1: home screen



Tools/Software Used

1. Figma

2. Android Studio

3. Flutter SDK

Personal Reflection

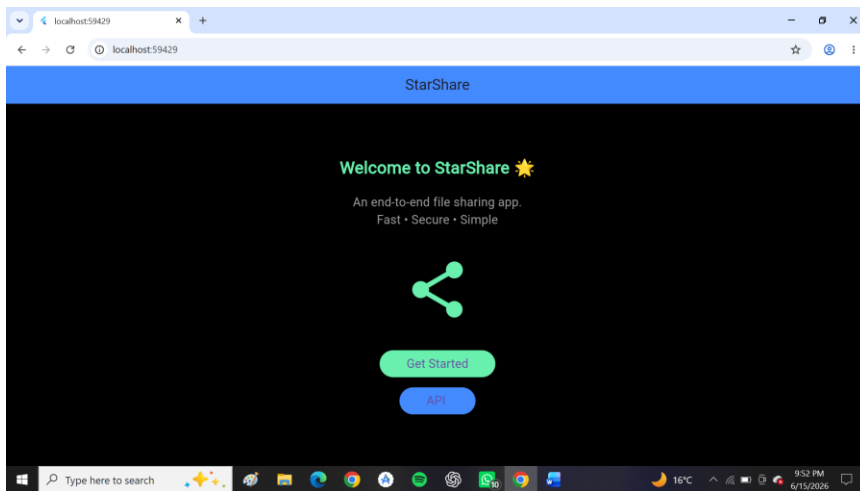
I refined the user interface design and improved consistency across all screens. The finalized wireframes provided a clear development roadmap and enhanced the overall usability of the StarShare application.

Week 4: Event Handling and User Interaction

Practical Activities Undertaken

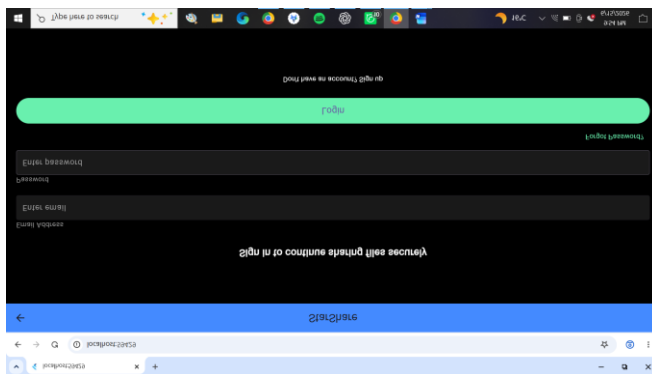
1. Implemented button click events in Flutter UI.
2. Added navigation between screens using routes.
3. Implemented form validation for login and registration.
4. Handled user inputs using TextEditingController.
5. Added loading indicators during user interactions.
6. Implemented basic error handling for invalid inputs.

Figure1



The get started button leads you to onboarding page.

Figure 2



The login button takes you to home page.

Tools used:

1. Flutter SDK
2. Android Studio
3. Dart Programming Language

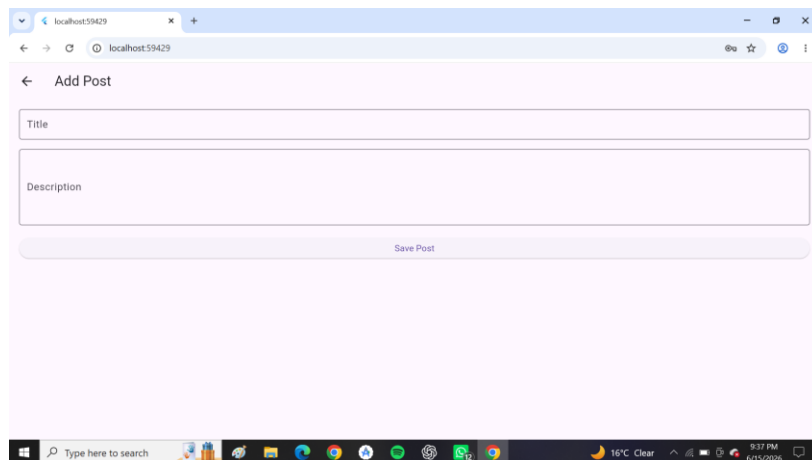
Personal Reflection

I learned how to handle user interactions in Flutter using event listeners, navigation, and form validation. This improved the responsiveness and usability of the application.

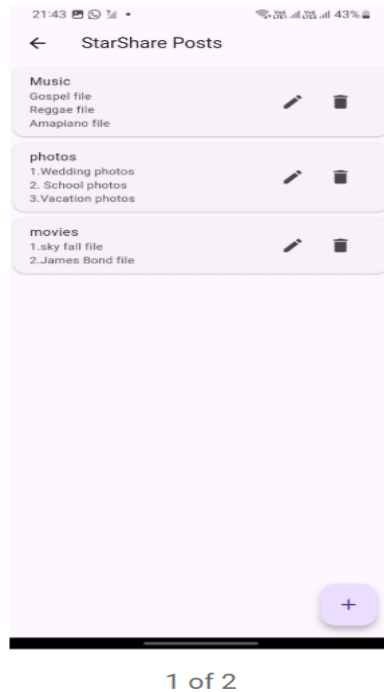
Week 5: Data Management

Practical Activities Undertaken

1. Created data models for users and posts.
2. Implemented local state management using `setState()`.
3. Integrated API service for data fetching and posting.
4. Parsed JSON responses from backend.
5. Displayed dynamic data using `ListView.builder`.
6. Handled loading and error states in data rendering.



Allows adding and saving of posts



This one shows the editing and the deleting part of the system.

Tools Used

1. Flutter SDK
2. Dart
3. API Service (HTTP package)
4. Android Studio

Personal Reflection

I gained practical experience in managing and displaying dynamic data in Flutter applications. I also learned how to integrate backend APIs and handle JSON data effectively.

Week 6: Networking and APIs

Practical Activities Undertaken

1. Created REST API service for backend communication.
2. Implemented HTTP GET and POST requests in Flutter.
3. Connected Flutter application to backend server.
4. Handled API responses and errors.
5. Implemented asynchronous programming using `async/await`.
6. Tested network connectivity and data flow.

Figure1

This one shows a button for API for backend communication.

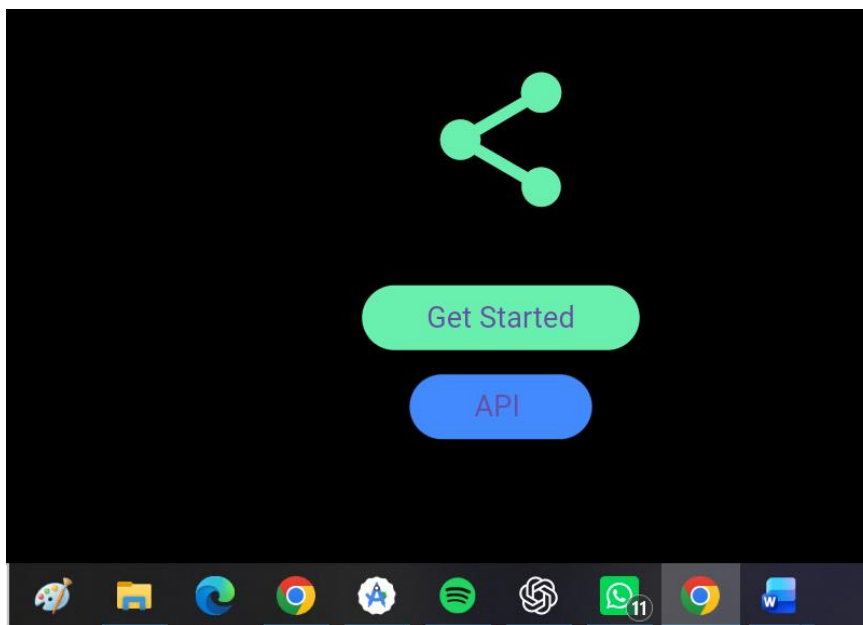


Figure 2

This shows when the internet access is denied, the server cannot be reached.

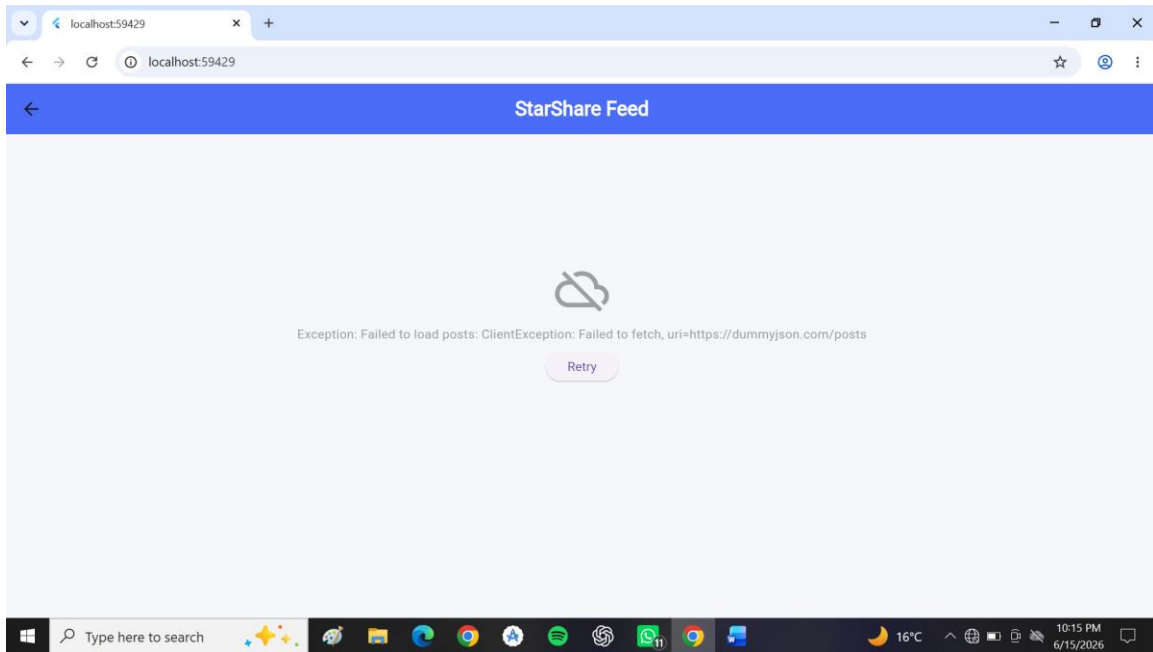
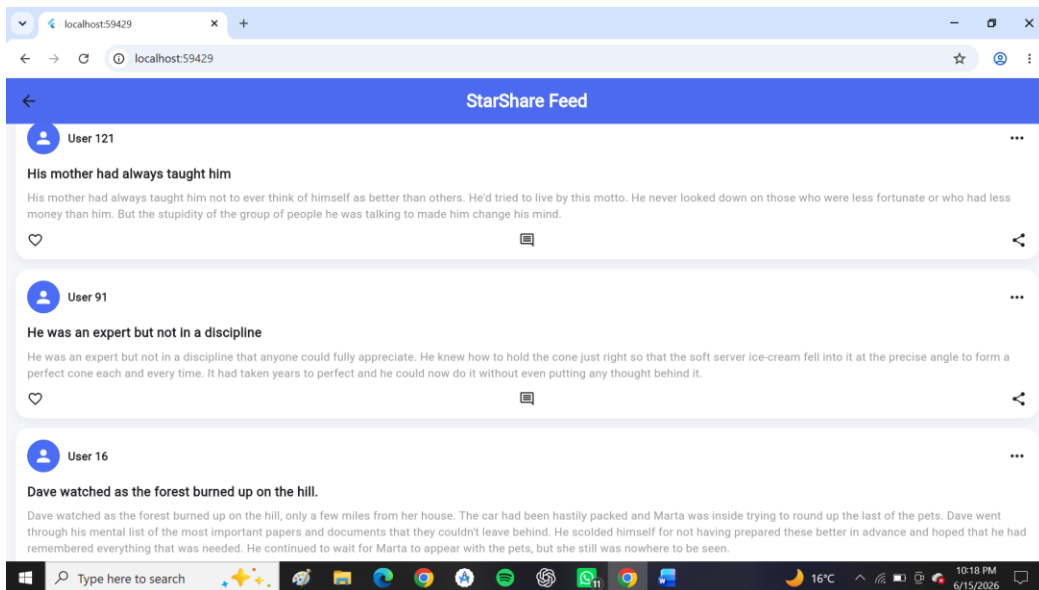


Figure 3

1. Handled API responses and the internet error and reached the information.



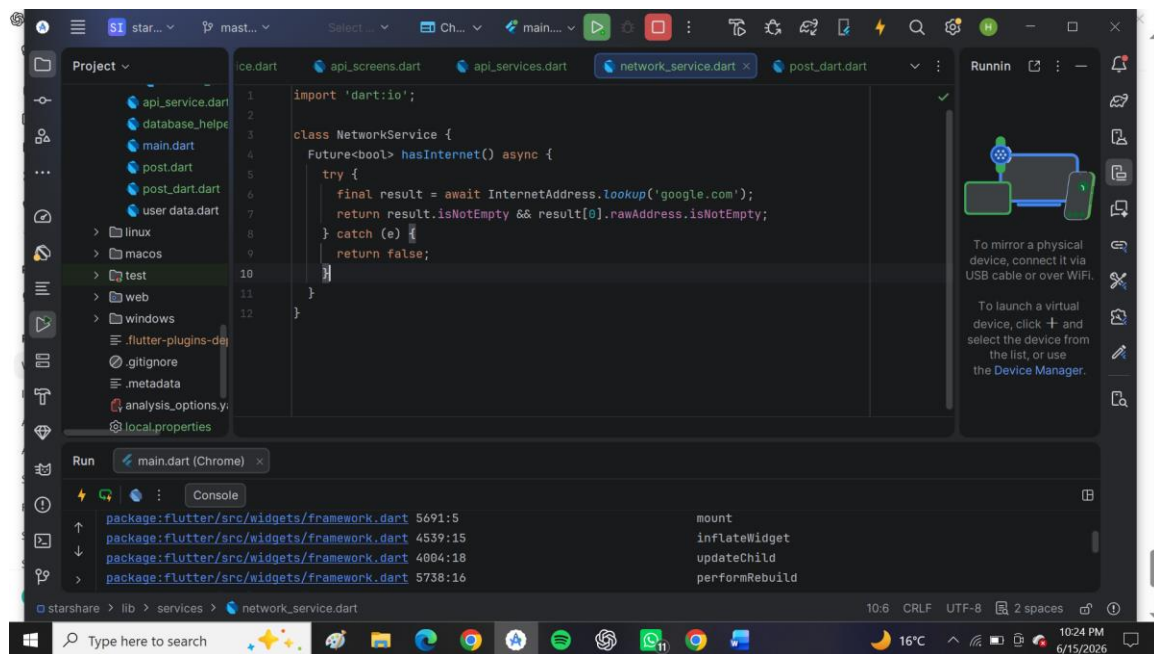
Tools

1. Flutter SDK
2. Dart
3. HTTP package
4. Backend API (Node.js / PHP / Firebase if used)
5. Android Studio

Personal Reflection

I learned how to connect a mobile application to external APIs and manage real-time data exchange. This improved my understanding of networking, asynchronous programming, and backend integration.

The network connection string code



Why does my system (Starshare) require networking?

StarShare requires networking to upload, sync, and retrieve files and user data between devices and the server.

Week 7: Application Architecture

Same as week 1

Week 8: Project Planning

Week 9: Device Features and Sensors

Same as week 1

Week 10: Notifications and Background Services

Same as week 1

Week 11: Mobile Security

Same as week 1

Week 12: Testing and Debugging

Same as week 1

Week 13: Application Deployment

Same as week 1

Week 14: Final Project Presentation

Same as week 1

Semester Project Tracking

Project Title

Project Objectives

Project Requirements

Development Milestones

Testing Results

Lessons Learned

Future Improvements

End of Semester Reflection

Summary of Skills Acquired

Most Interesting Topic

Major Challenges Faced

How Challenges Were Overcome

Professional Growth Achieved

Overall Course Experience

Final Approval

Student Name: Haron Mutai

Student Signature:

Date: 9th June 2026

Lecturer Name: Michael Nyoro

Lecturer Remarks: _____

Date: _____